

OOP Day07 Assignment

Task 1 – Abstract Class (Payment System)

Requirements:

1. Class Design

- Create an abstract class named PaymentMethod.
- Derived classes:
 - CreditCardPayment
 - PayPalPayment
 - BankTransferPayment

2. Abstract Base Class: PaymentMethod

Properties:

- TransactionId (int)
- Amount (decimal)

Methods:

- abstract bool ProcessPayment()
- virtual void DisplayInfo()

3. Derived Classes

CreditCardPayment:

- Properties: CardNumber, CardHolderName
- Override ProcessPayment()

PayPalPayment:

- Property: Email
- Override ProcessPayment()

BankTransferPayment:

- Properties: BankName, AccountNumber
- Override ProcessPayment()
- Override DisplayInfo ()

4. Polymorphism Requirement

- Use a PaymentMethod reference.
- Assign different payment objects.
- Call ProcessPayment() dynamically.

5. Output Expectations

- Each payment type processes differently.
- Abstraction and polymorphism should be demonstrated.

Task 2 – Static Class + Visitor Counter

Requirements:

1. Static Class Design

- Create a static class named VisitorCounter.

2. Static Class: VisitorCounter

Static Properties:

- TotalVisitors (int)

Static Constructor:

- Initializes TotalVisitors to 0 and runs once before first use.

Static Methods:

- Increase()
- DisplayCount()

3. Visitor Class

- Regular class with a parameterized constructor that increments TotalVisitors each time a new visitor object is created.

4. Output Expectations

- Demonstrates static constructor and shared static data.
- Shows global visitor counting behavior.